

NETWORK ANALYSIS And SURVEYING For 5G Wi-Fi



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The Digital India initiative centres broadly around creating a digitally super-empowered India. This is estimated to cost ₹ 1.13 trillion and would allow India to provide high-speed Internet services and offer manifold services in real-time for mutually-online and mobile platforms.

In the first phase, the initiative aims to supply free Wi-Fi across rail infrastructure and provide broader connectivity to the outer reaches of India. With BSNL committing ₹ 60 billion to make available 40,000 Wi-Fi hotspots in the country, there is renewed emphasis on bringing unsurpassed Wi-Fi technologies and solutions to the sub-continent.

With the support of the government of India, global companies like Facebook and Google are also supporting Digital India initiative and helping in bringing connectivity to the nation.

Wi-Fi technologies and solutions would play a significant role in making this programme a reality, as it is an integral and crucial part of extending connectivity all the way through a district. It is not being done by wires or through communication channels, but through a Cloud network. The architecture is simple, where we can

ascertain its functions, nodes and different types of services and applications. Fig. 1 shows the various Cloud network services.

Applications can be built on topologies such as mesh and micromesh as both modes (ad-hoc and static) could be used for the long and unreachable terminals.

Fig. 2 shows a wireless router connecting the online server, where Cloud services are available. The services are implemented from the wireless sensor node data and covers the users available in the network. The distributed scheme of available bandwidth among users integrates the district or state by the Cloud network and other Wi-Fi solutions such as the Internet of Things (IoT) and Fiber Everywhere for roaming users.

Setting up operating modes of Wi-Fi

Let us see how Wi-Fi is set up in desktop computers or laptops that run on Windows 7 platform, and how number of users, channels and signal strength is determined by a simulator, followed by MATLAB solution for signal-to-noise ratio. Connection made by the user in the platform requires evaluation and troubleshooting for uninterrupted high-speed service.

Under Device Manager, 802.11 pream-

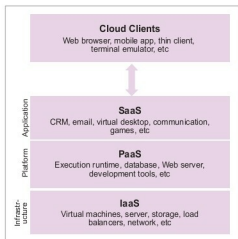


Fig. 1: Cloud network services

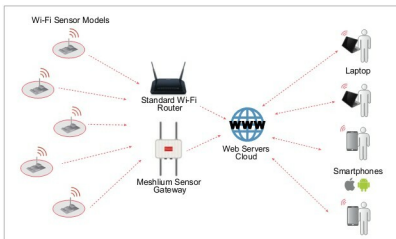


Fig. 2: Wi-Fi mesh network Cloud